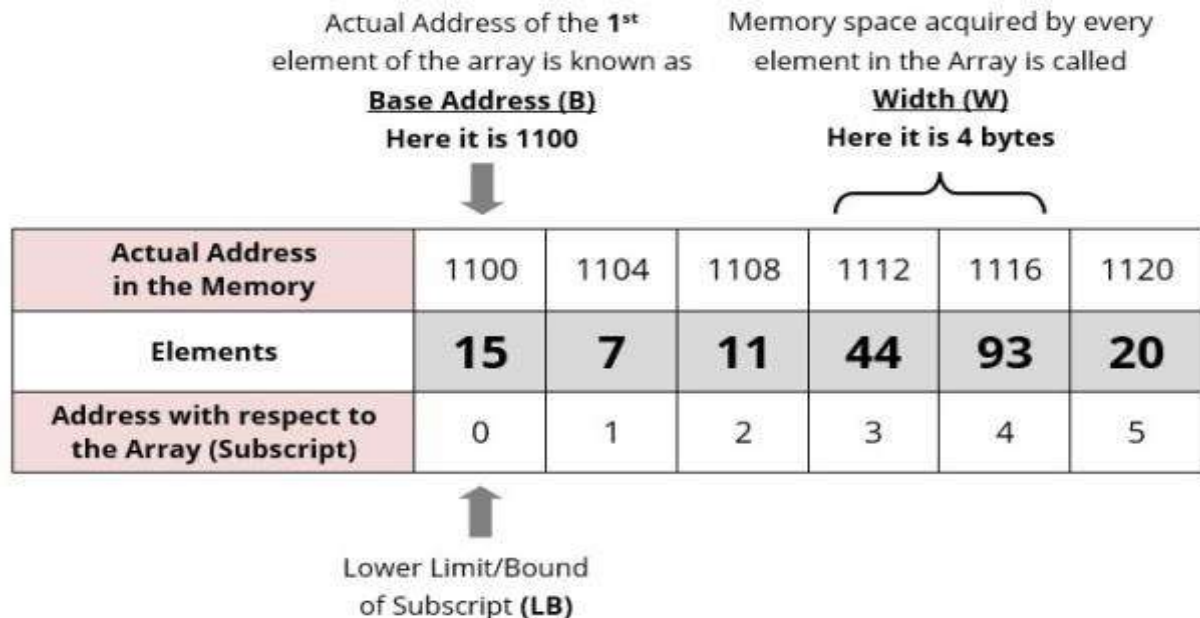


Memory Address Calculation in an Array

Address Calculation in single (one) Dimension Array:



Address of an element of an array say "A[I]" is calculated using the following formula:

$$\text{Address of A [I]} = B + W * (I - LB)$$

Where,

B = Base address

W = Storage Size of one element stored in the array (in byte)

I = Subscript of element whose address is to be found

LB = Lower limit / Lower Bound of subscript, if not specified assume 0 (zero)

Example:

Given the base address of an array **B[1300.....1900]** as 1020 and size of each element is 2 bytes in the memory. Find the address of **B[1700]**.

Solution:

The given values are: B = 1020, LB = 1300, W = 2, I = 1700

$$\text{Address of A [I]} = B + W * (I - LB)$$

$$= 1020 + 2 * (1700 - 1300)$$

$$= 1020 + 2 * 400$$

$$= 1020 + 800$$

$$= 1820 \text{ [Ans]}$$

Address Calculation in Double (Two) Dimensional Array:

While storing the elements of a 2-D array in memory, these are allocated contiguous memory locations. Therefore, a 2-D array must be linearized so as to enable their storage. There are two alternatives to achieve linearization: Row-Major and Column-Major.

		Column Index			
		0	1	2	3
Row Index	0	8	6	5	4
	1	2	1	9	7
	2	3	6	4	2

Two-Dimensional Array

Row-Major (Row Wise Arrangement)

8	6	5	4	2	1	9	7	3	6	4	2
Row 0				Row 1				Row 2			

Column-Major (Column Wise Arrangement)

8	2	3	6	1	6	5	9	4	4	7	2
Column 0			Column 1			Column 2			Column 3		

Address of an element of any array say “**A[I][J]**” is calculated in two forms as given:
(1) Row Major System (2) Column Major System

Row Major System:

The address of a location in Row Major System is calculated using the following formula:

$$\text{Address of A [I][J]} = B + W * [N * (I - L_r) + (J - L_c)]$$

Column Major System:

The address of a location in Column Major System is calculated using the following formula:

$$\text{Address of A [I][J] Column Major Wise} = B + W * [(I - L_r) + M * (J - L_c)]$$

Where,

B = Base address

I = Row subscript of element whose address is to be found

J = Column subscript of element whose address is to be found

W = Storage Size of one element stored in the array (in byte)

L_r = Lower limit of row/start row index of matrix, if not given assume 0 (zero)

L_c = Lower limit of column/start column index of matrix, if not given assume 0 (zero)

M = Number of row of the given matrix

N = Number of column of the given matrix

Important: Usually number of rows and columns of a matrix are given (like A[20][30] or A[40][60]) but if it is given as **A[L_r- - - - U_r, L_c- - - - U_c]**. In this case number of rows and columns are calculated using the following methods:

Number of rows (**M**) will be calculated as = **(U_r – L_r) + 1**

Number of columns (**N**) will be calculated as = **(U_c – L_c) + 1**

And rest of the process will remain same as per requirement (Row Major Wise or Column Major Wise).

Examples:

Q 1. An array X [-15.....10, 15.....40] requires one byte of storage. If beginning location is 1500 determine the location of X [15][20].

Solution:

As you see here the number of rows and columns are not given in the question. So they are calculated as:

Number of rows say **M = (U_r – L_r) + 1 = [10 – (- 15)] + 1 = 26**

Number of columns say **N = (U_c – L_c) + 1 = [40 – 15]] + 1 = 26**

(i) Column Major Wise Calculation of above equation

The given values are: $B = 1500$, $W = 1$ byte, $I = 15$, $J = 20$, $Lr = -15$, $Lc = 15$, $M = 26$

Address of $A[I][J] = B + W * [(I - Lr) + M * (J - Lc)]$

$$= 1500 + 1 * [(15 - (-15)) + 26 * (20 - 15)] = 1500 + 1 * [30 + 26 * 5] = 1500 + 1 * [160] = 1660 \text{ [Ans]}$$

(ii) Row Major Wise Calculation of above equation

The given values are: $B = 1500$, $W = 1$ byte, $I = 15$, $J = 20$, $Lr = -15$, $Lc = 15$, $N = 26$

Address of $A[I][J] = B + W * [N * (I - Lr) + (J - Lc)]$

$$\begin{aligned} &= 1500 + 1 * [26 * (15 - (-15)) + (20 - 15)] = 1500 + 1 * [26 * 30 + 5] = 1500 + 1 * [780 + 5] = 1500 + 785 \\ &= 2285 \text{ [Ans]} \end{aligned}$$